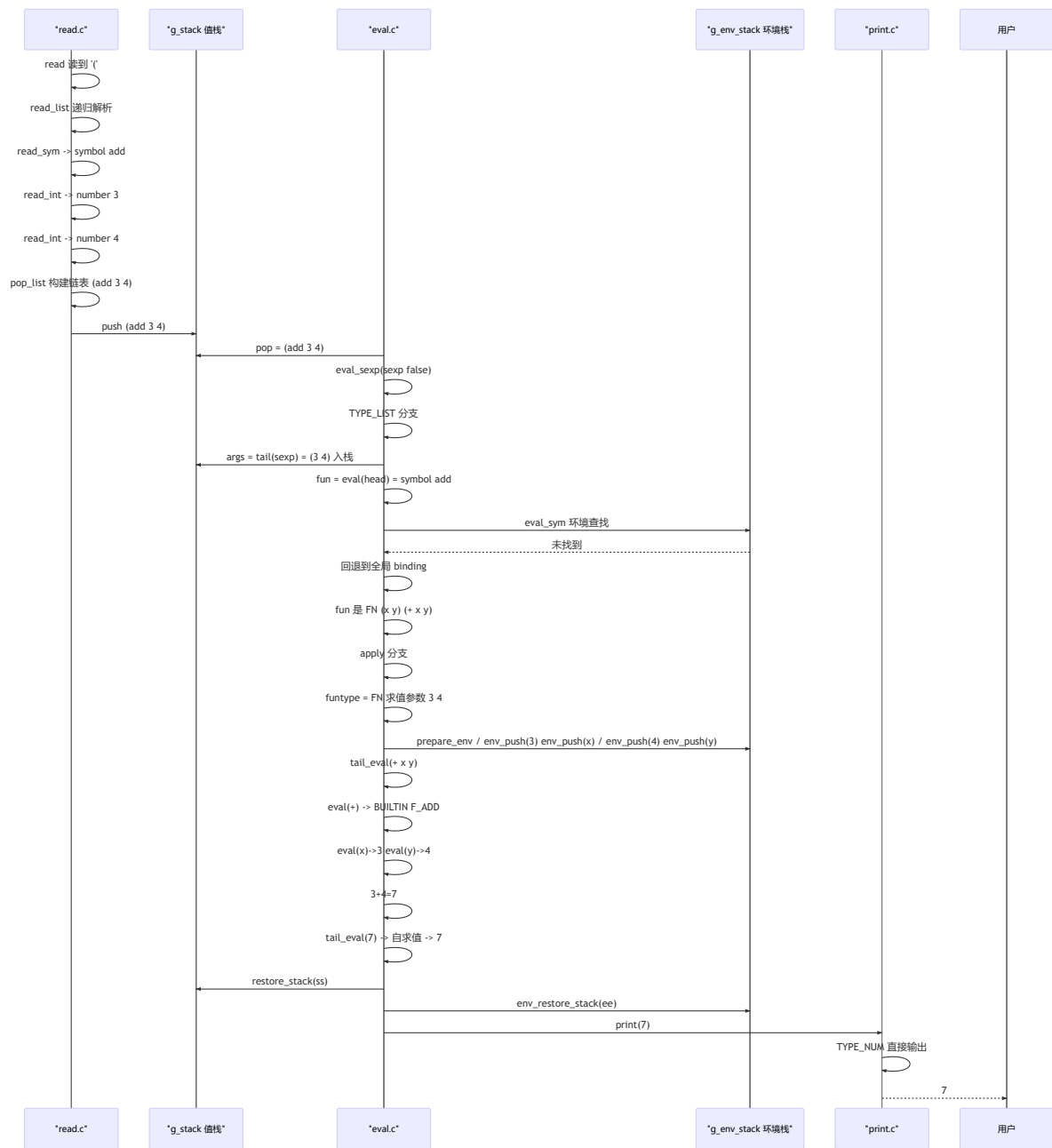
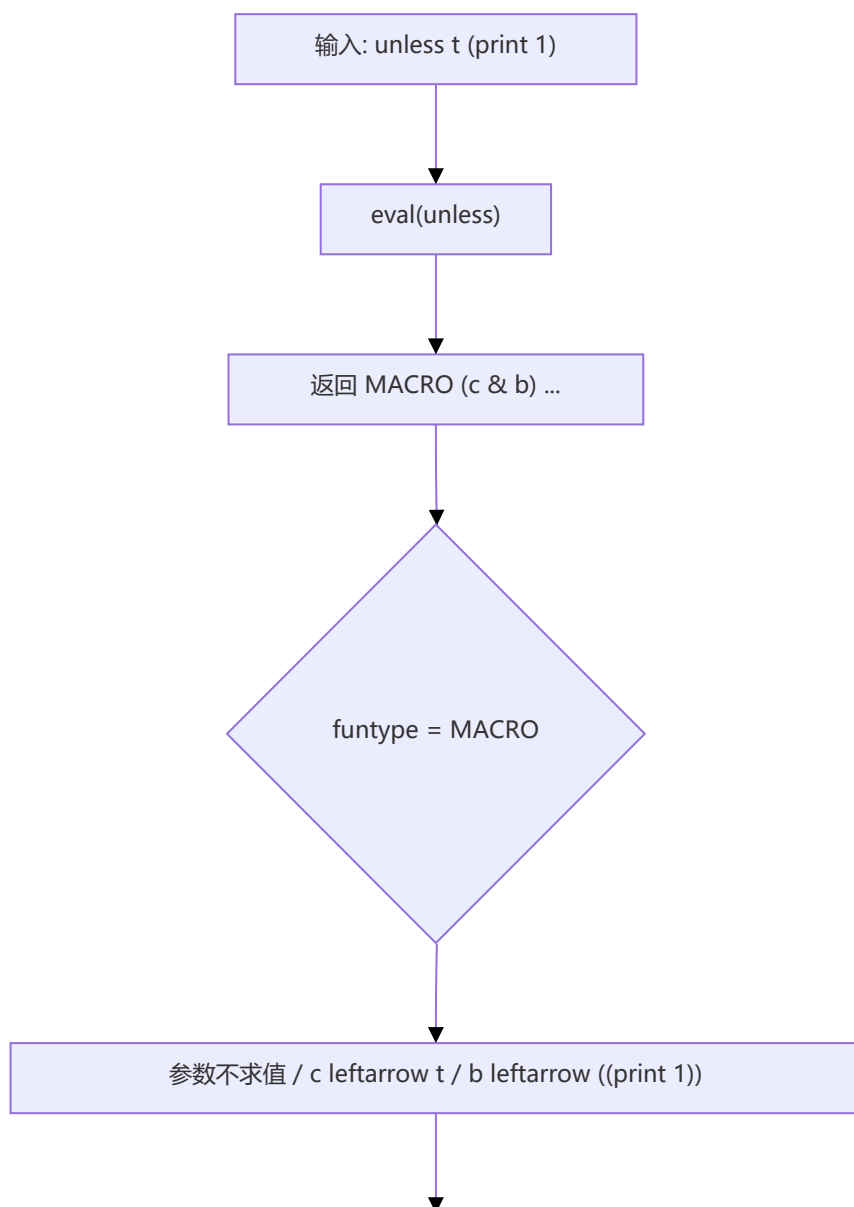


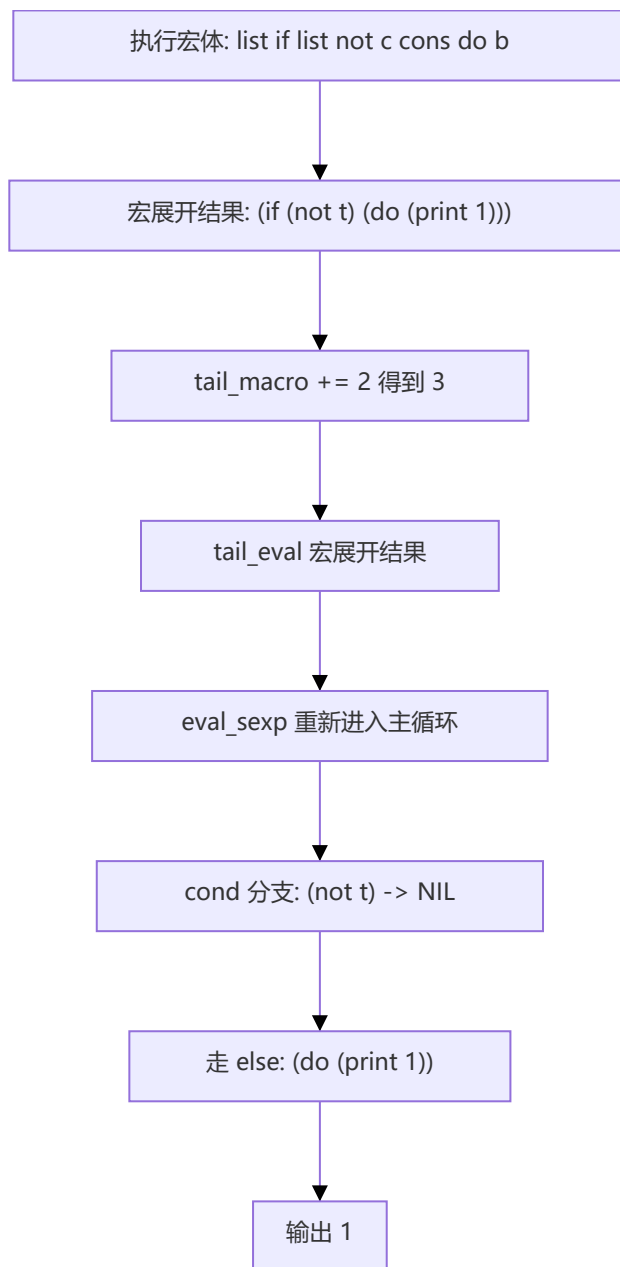
1. 顶层架构图





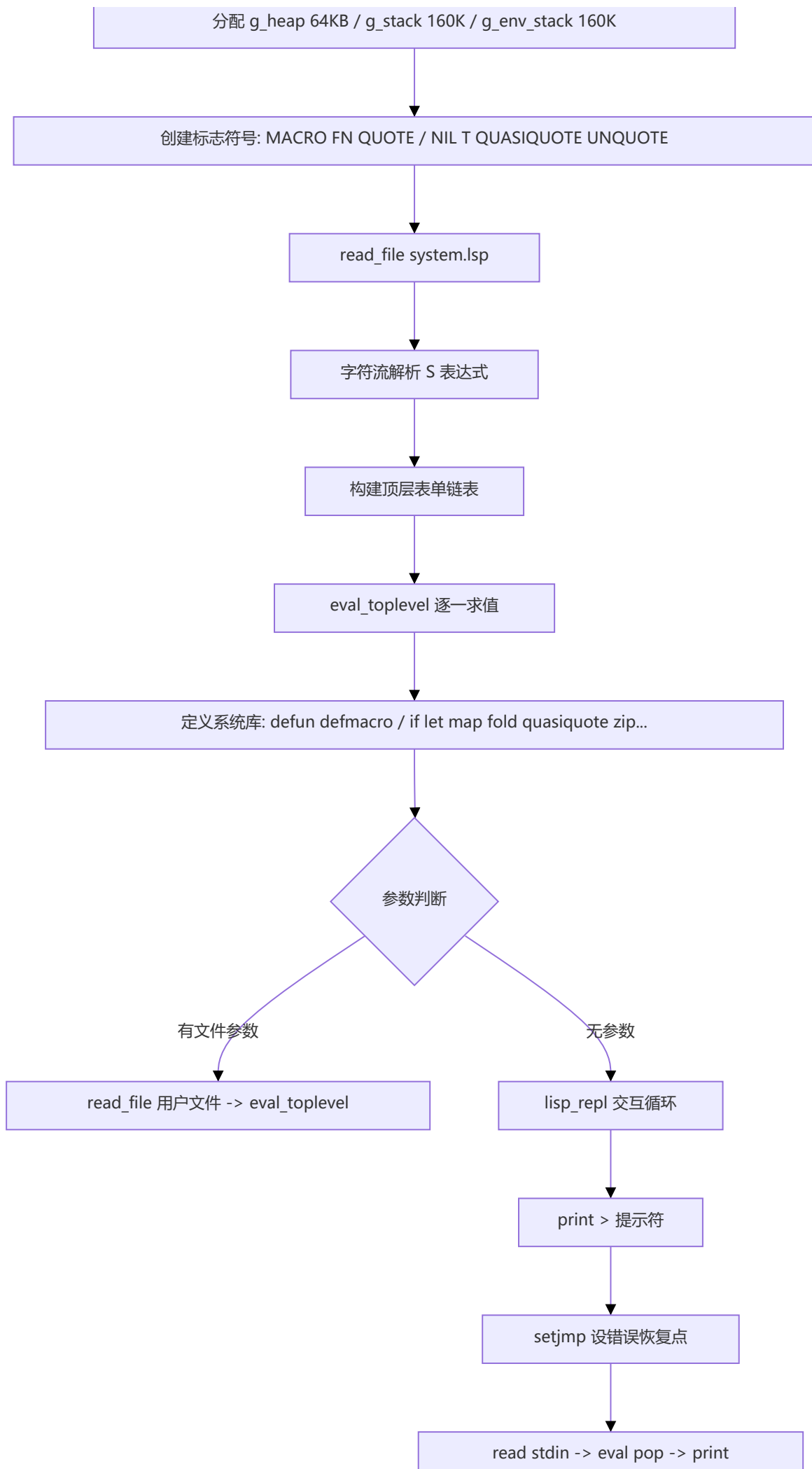
4. 宏展开流程 — (unless #t (print 1))





5. 闭包捕获流程 — (let ((x 42)) (fn (y) (+ x y)))

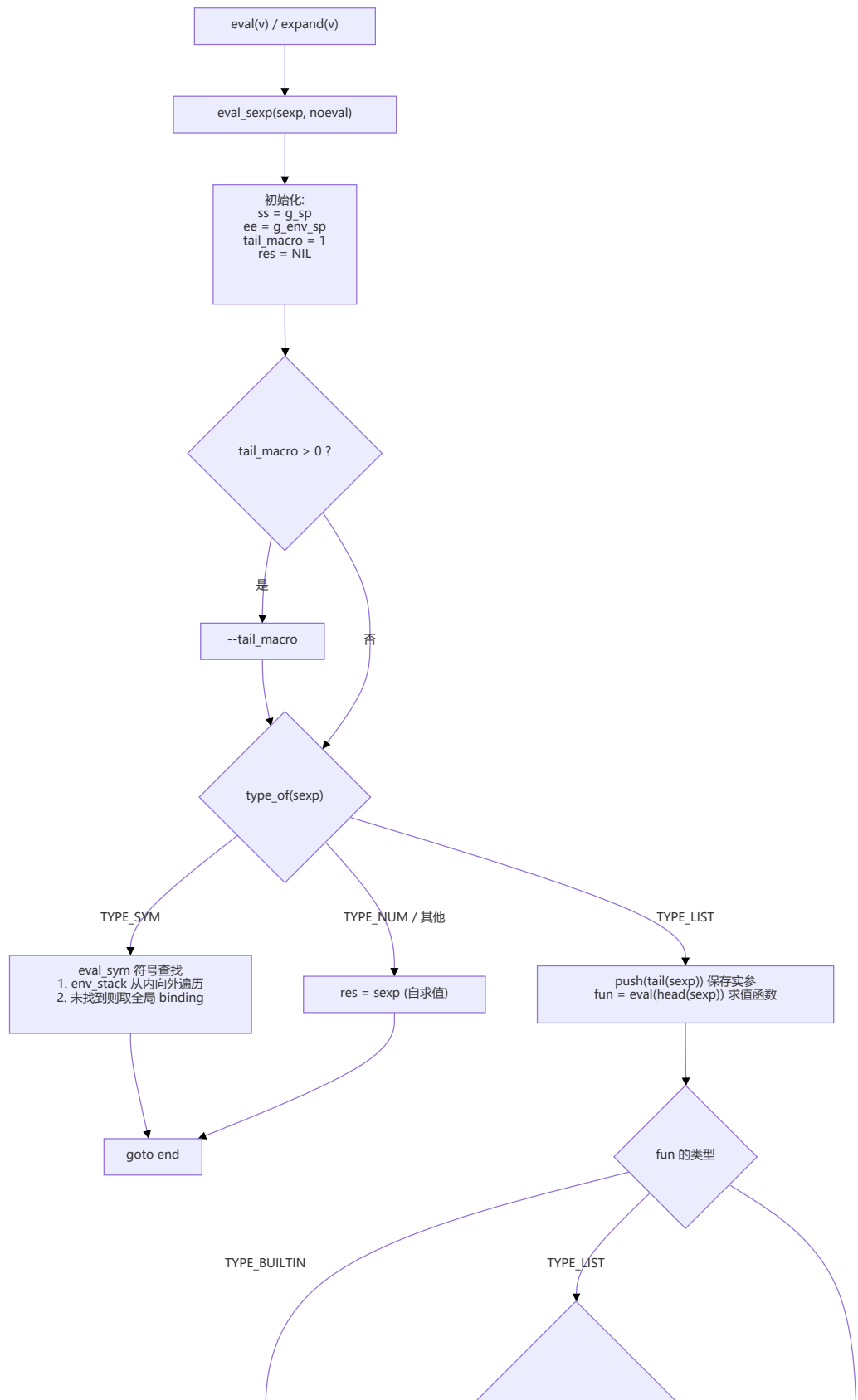


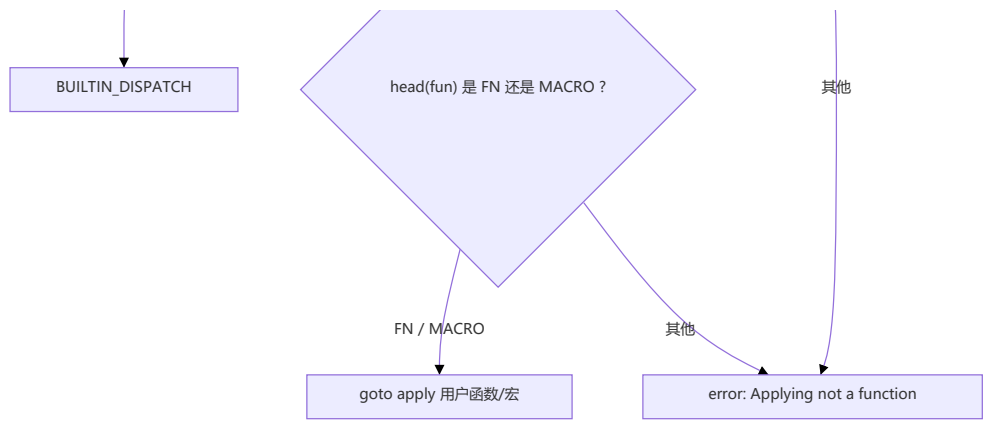




8. eval 详细流程图

8.1 eval_sexp 主循环完整结构

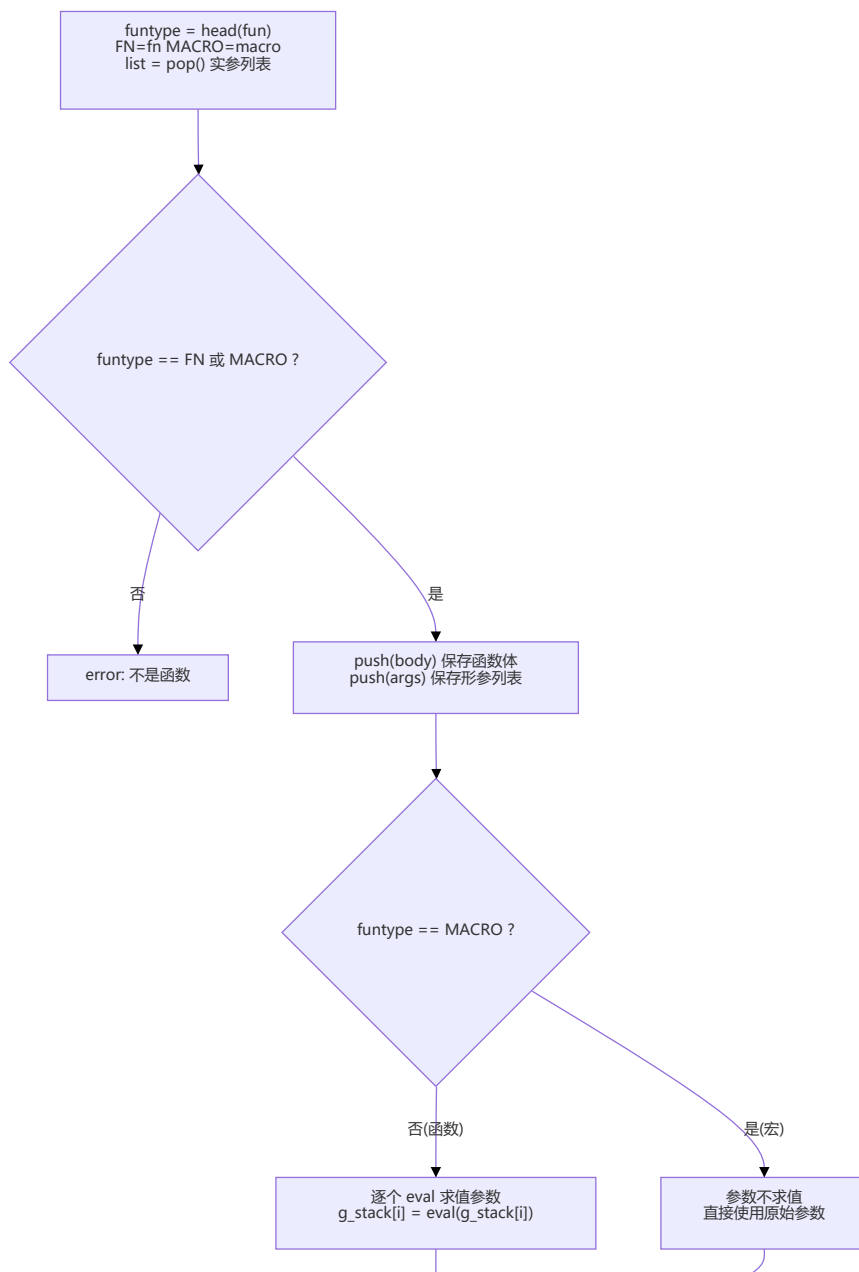


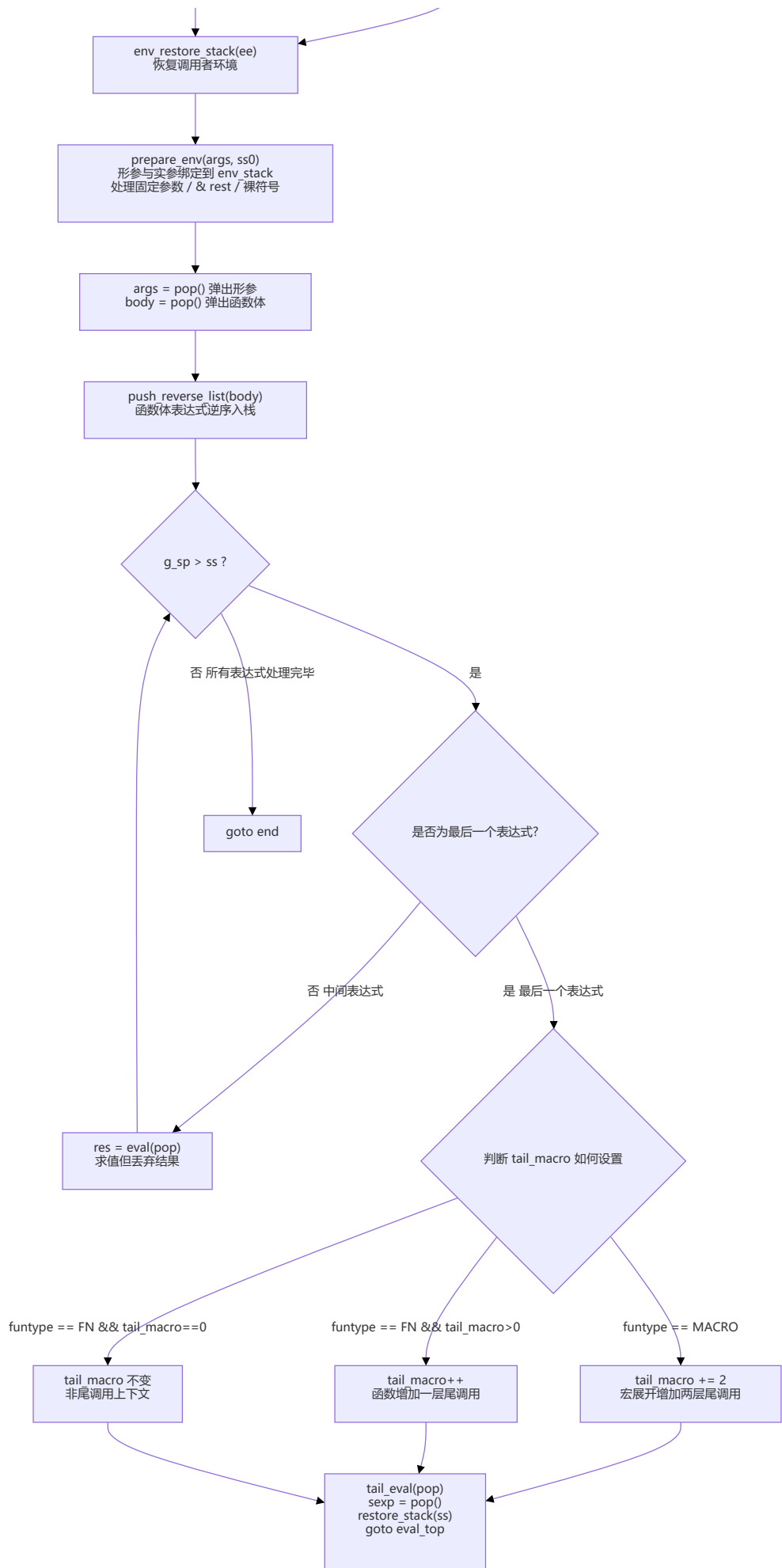


8.2 builtin 分发与特殊形式

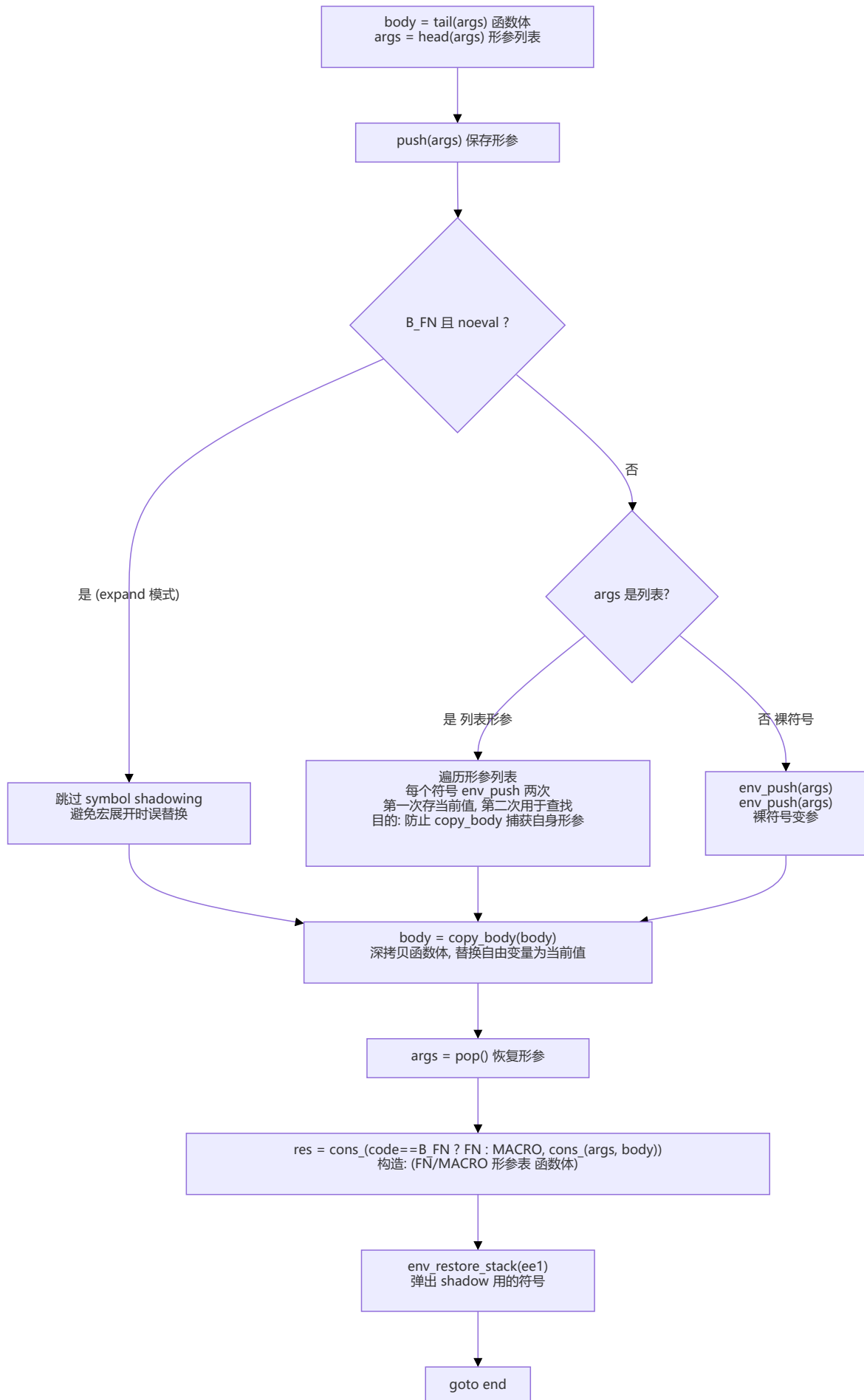


8.3 用户函数/宏应用 (apply)

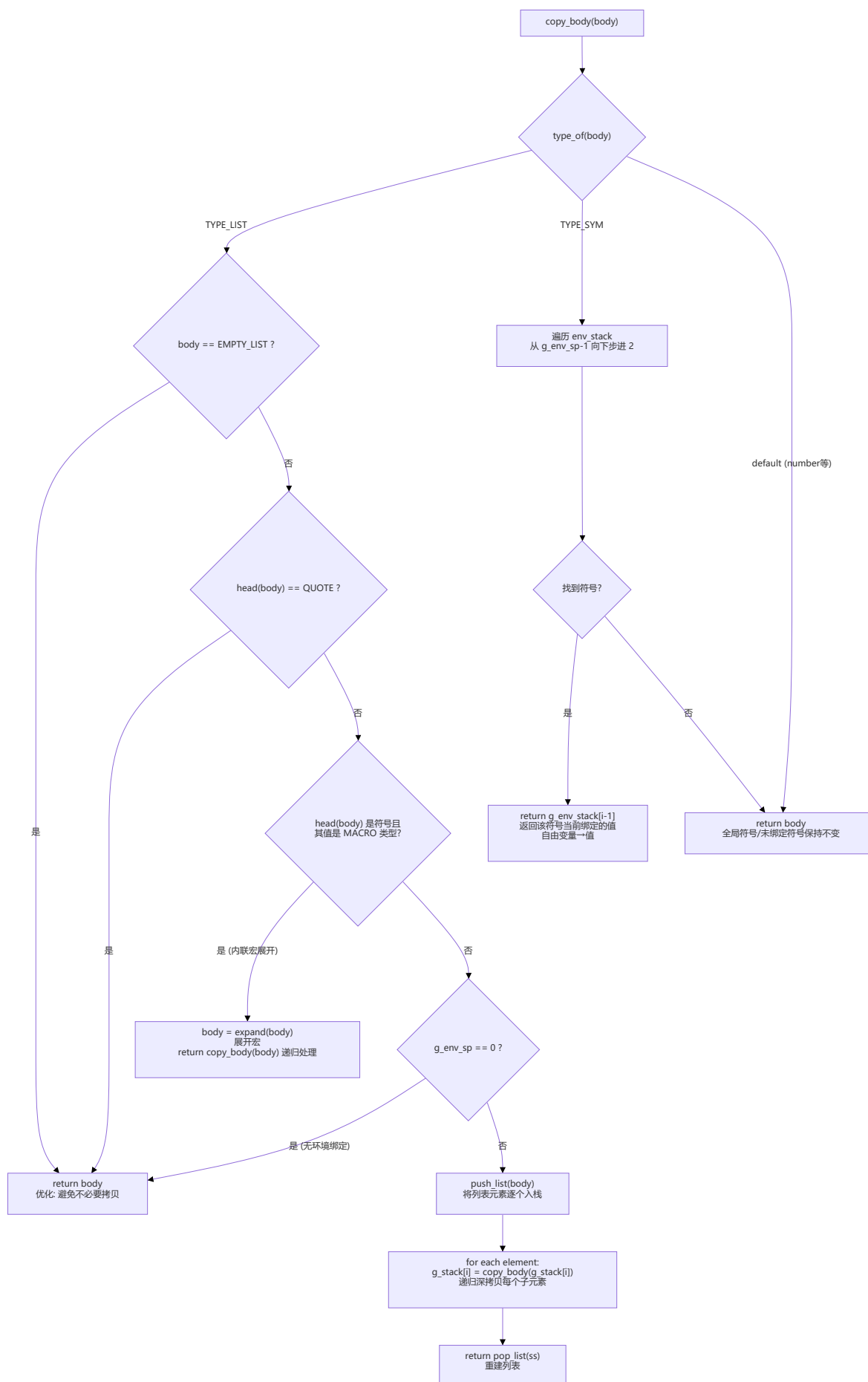




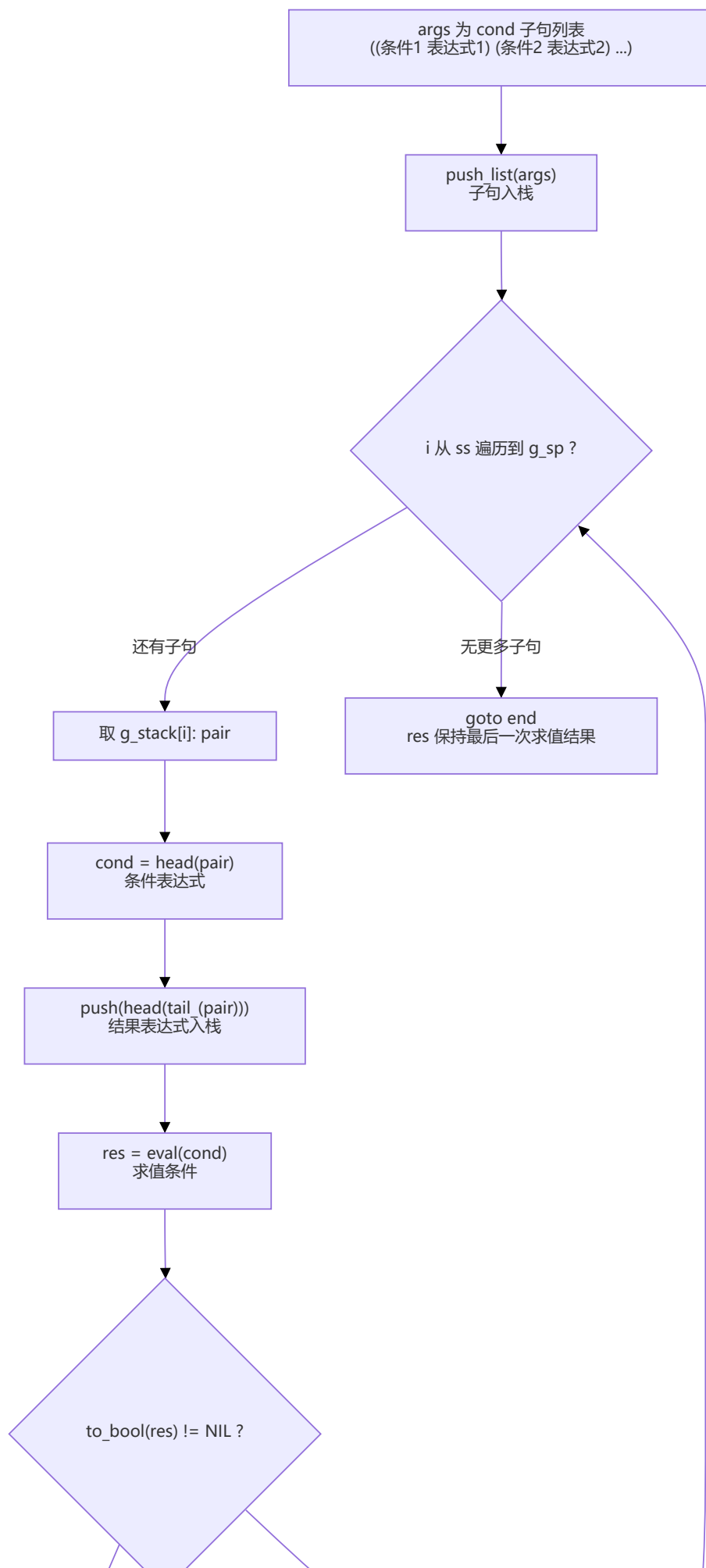
8.4 闭包创建 (B_FN / B_MACRO)

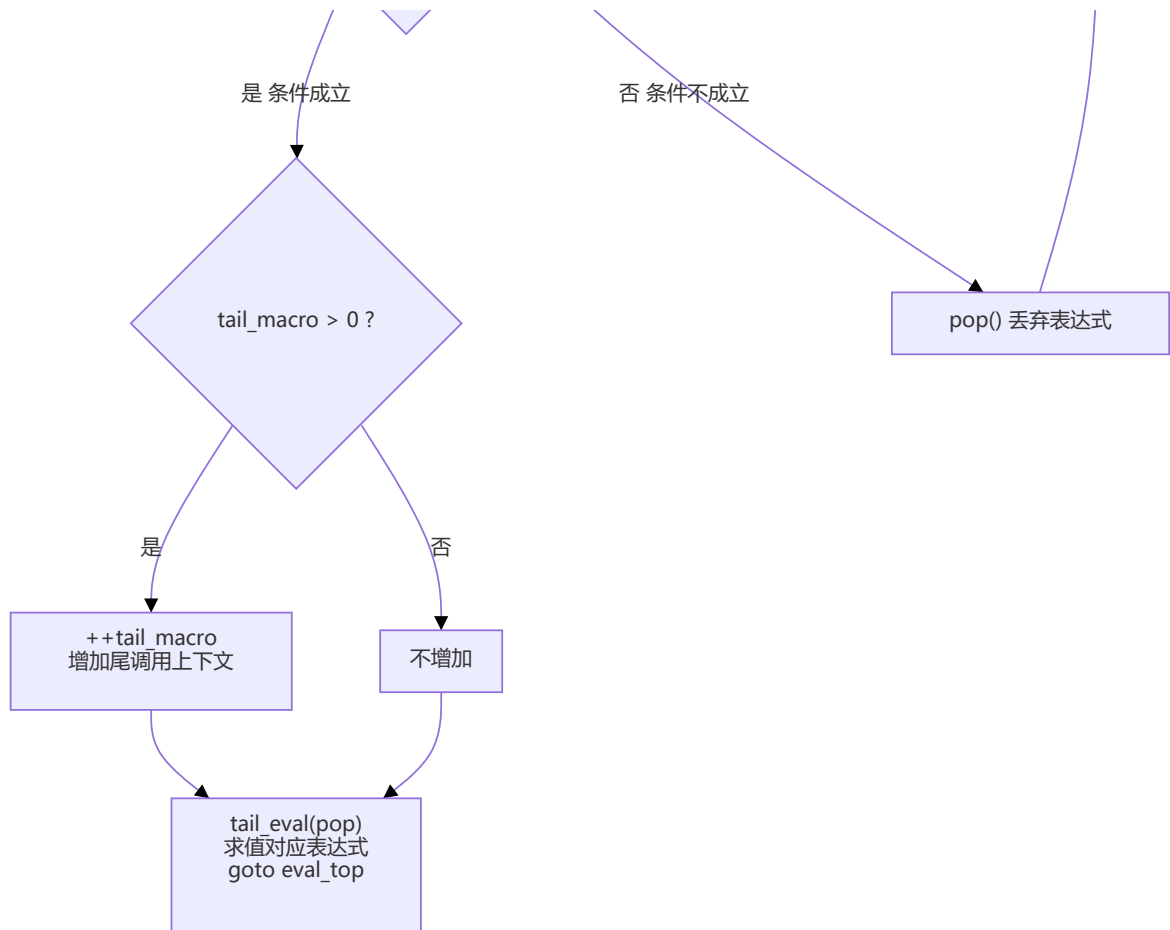


8.5 copy_body 详细流程 (深拷贝+闭包捕获)

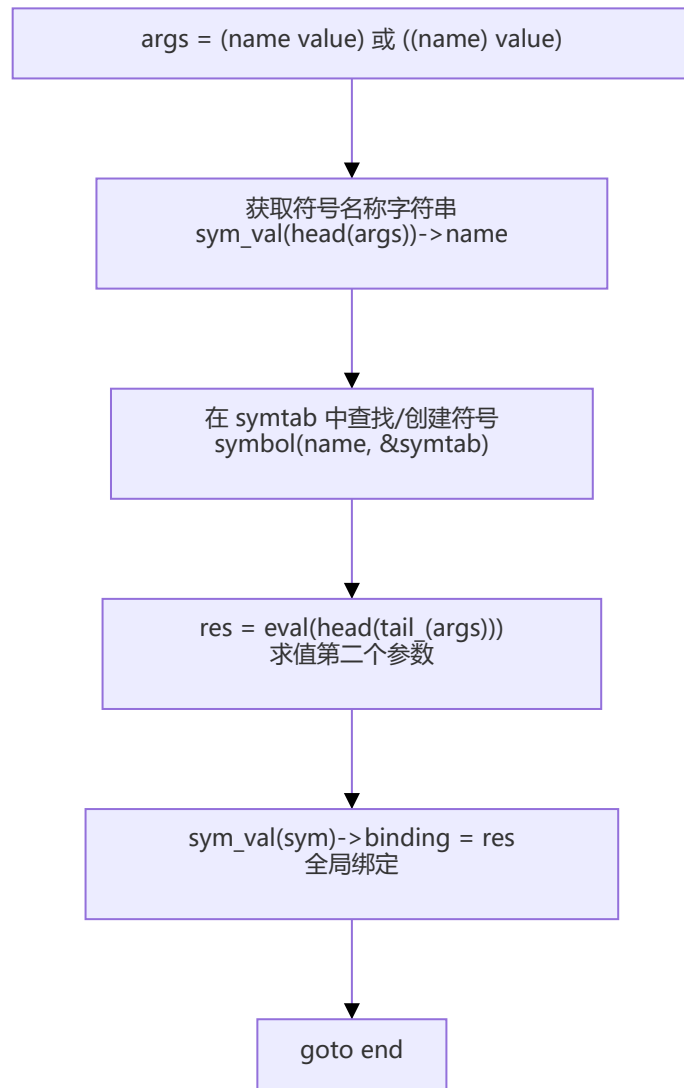


8.6 cond 特殊形式详解

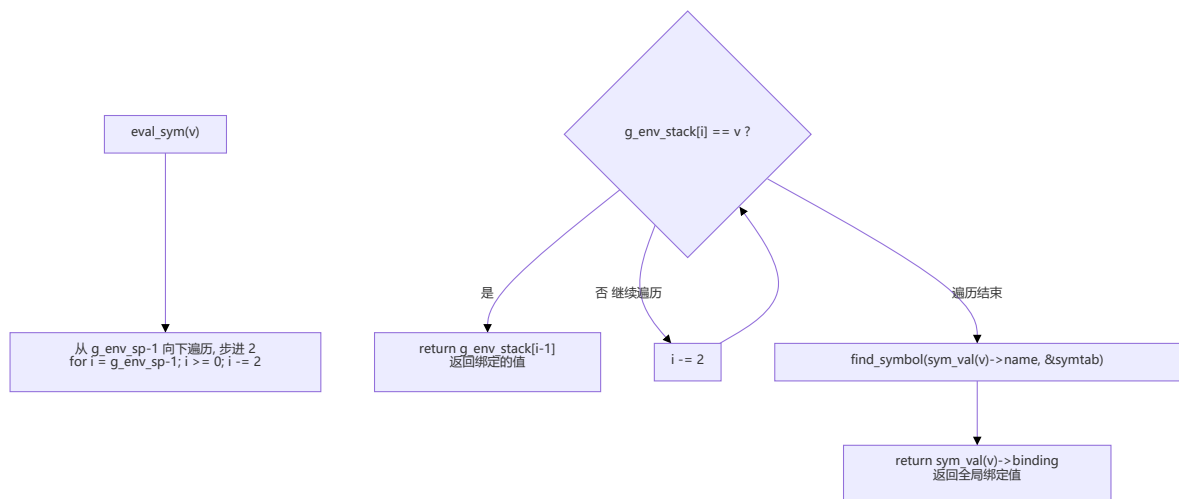




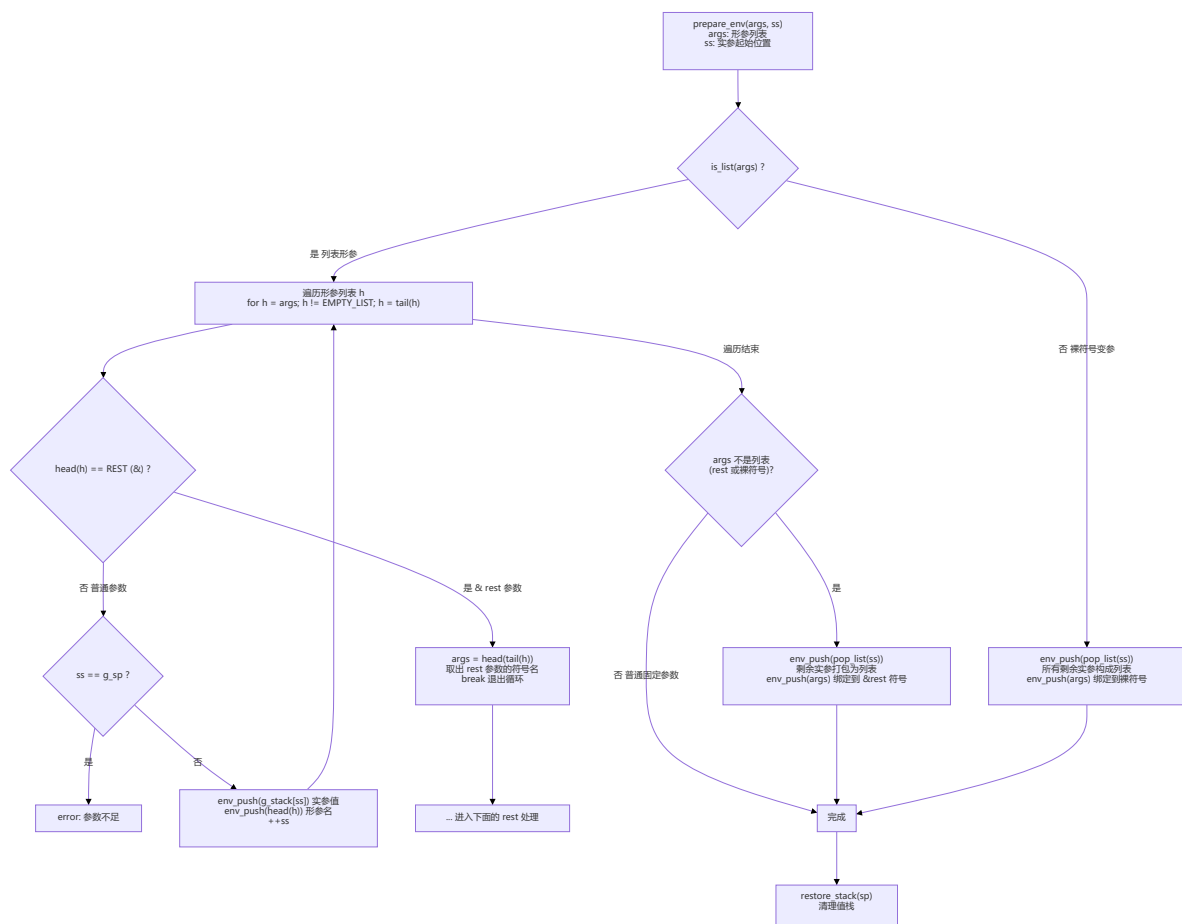
8.7 def 全局定义详解



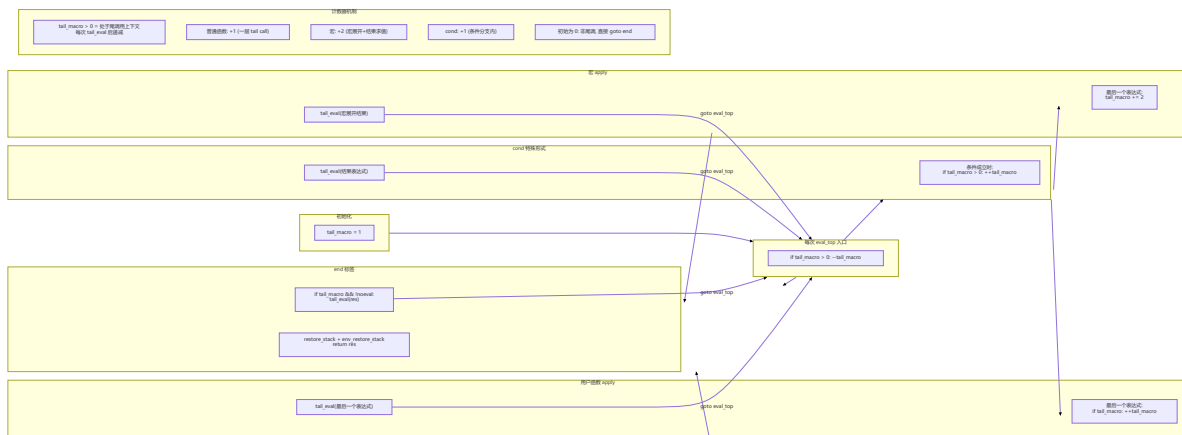
8.8 eval_sym 符号查找详细流程



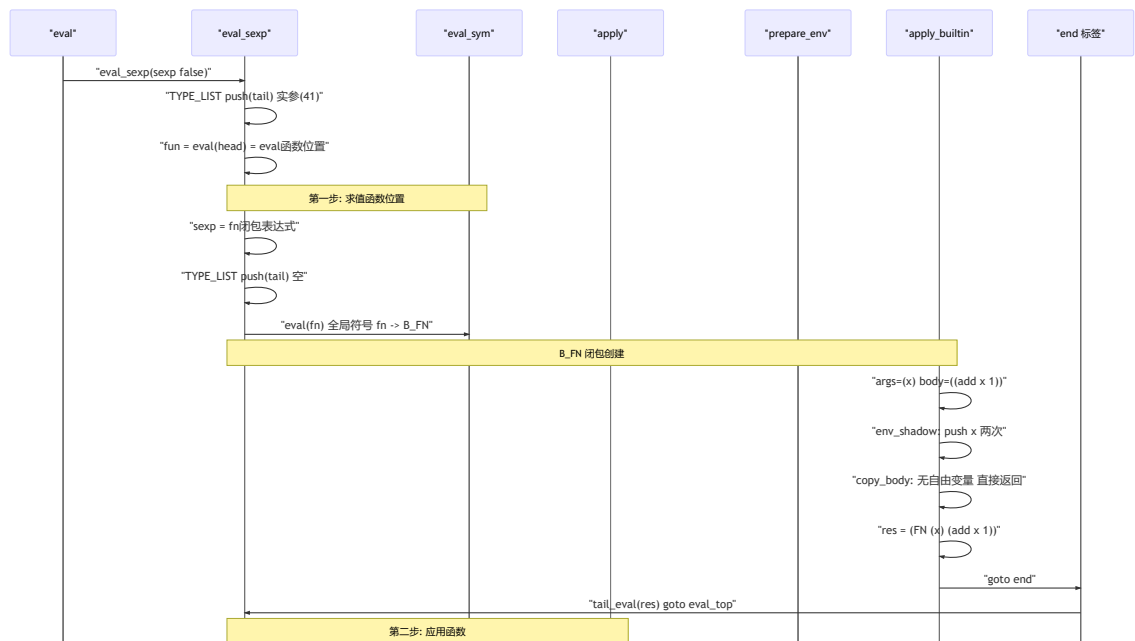
8.9 prepare_env 参数绑定详细流程

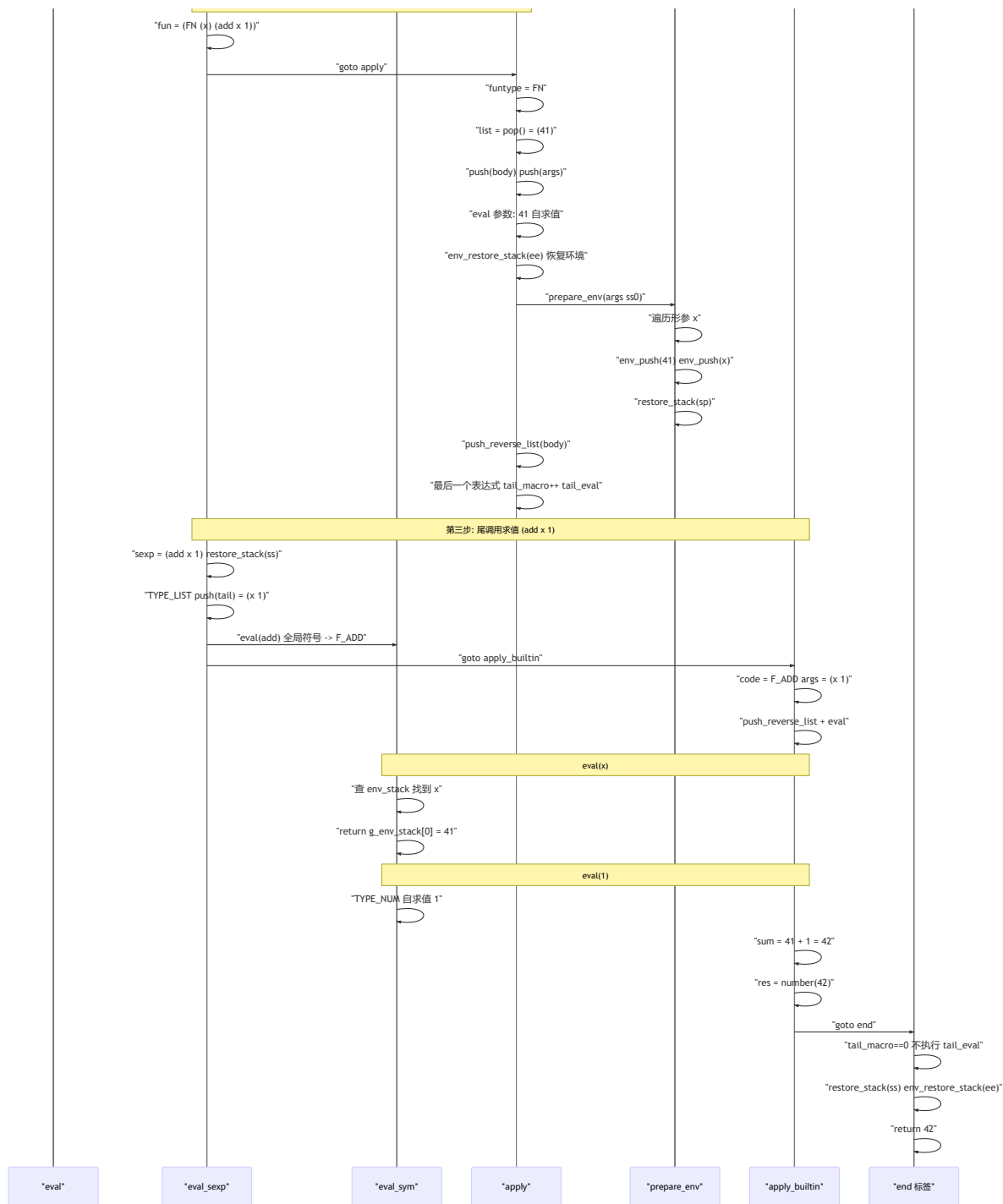


8.10 tail_macro 生命周期管理

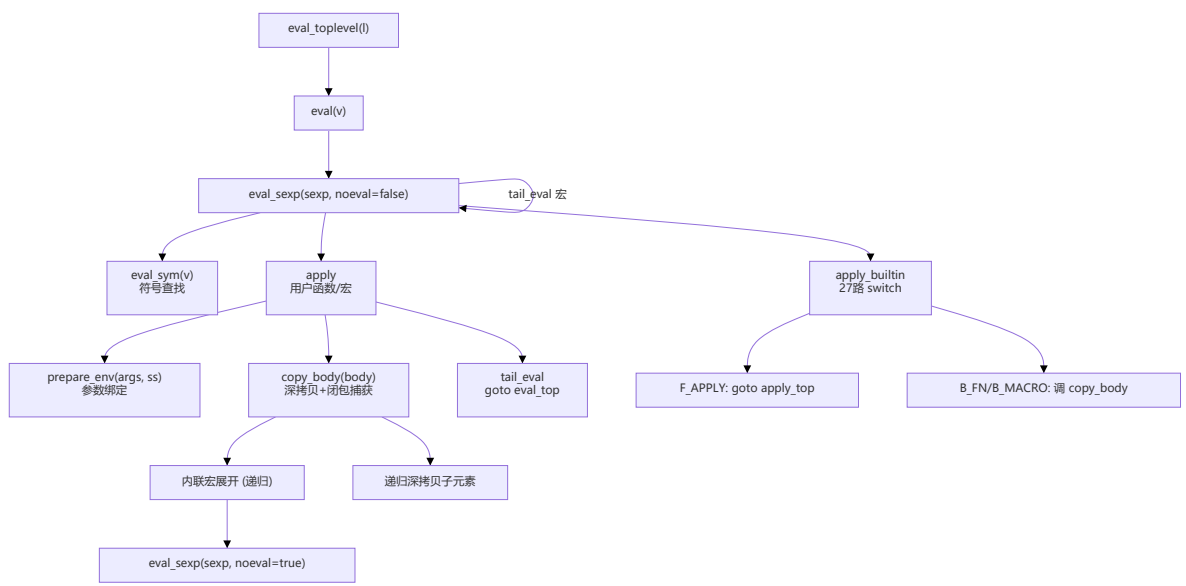


8.11 完整求值链路示例 — ((fn (x) (add x 1)) 41)





8.12 eval 模块函数依赖图



关键设计要点

阶段	核心文件	关键机制	数据格式
读取	<code>lisp_read.c</code>	递归下降解析器, 语法糖展开	字符流 -> tagged value (AST)
求值	<code>lisp_eval.c</code>	goto 主循环, 尾调用优化, 深拷贝闭包	tagged value -> tagged value
环境	<code>lisp_eval.c</code>	环境栈交替存储, prepare_env 绑定	value/symbol 配对栈
内置	<code>lisp_eval.c</code>	X-macro 枚举27个, switch 分发	Builtin {type code}
打印	<code>lisp_print.c</code>	类型分发, 递归列表打印	tagged value -> 字符串
GC	<code>lisp_core.c</code>	Cheney 停止复制, heap *1.5 扩容	from-space -> to-space
符号	<code>lisp_symbol.c</code>	BST 二叉搜索树, 哈希键	Symbol 节点